

3 OCEAN-ATMOSPHERE INTERACTIONS

IN CHAPTERS 1 AND 2, we discussed the atmosphere and the ocean separately. Each moves substantial amounts of energy around the planet and accounts for important characteristics of weather and climate. Interactions between the ocean and the atmosphere are coupled, meaning that the state of the ocean affects the state of the atmosphere and vice versa. Consequently, *coupled ocean-atmosphere interactions* create regional fluctuations in the circulation patterns of each, resulting in oscillatory behaviors of climate that can vary over weeks to decades and on local to global scales. These natural rhythms are the primary contributors to the *internal variability* of the climate system, and they persist absent any variation in external influences, such as the buildup of greenhouse gases in the atmosphere. In fact, they complicate the distinction between human- and naturally induced climate changes, although the evidence is quite clear that anthropogenic activities are now rapidly changing our climate.

In this chapter, we first consider the basic processes connecting the ocean and the atmosphere. We explore how energy, mass, and momentum move back and forth between the two and the implications of those exchanges on regional and global scales. We then investigate several important climate phenomena that either come about as a consequence of ocean-atmosphere interactions or represent other important kinds of oscillations within the climate system. The quintessential example of ocean-atmosphere coupling is the system that produces El Niño and La Niña, a phenomenon that we study in detail.

Exchanges at the Ocean-Atmosphere Interface

Because the ocean is vastly denser than the atmosphere (by ≈ 800 times), the interface between the two is stable. The ocean and the atmosphere nonetheless interact, and the principal way they do so is by the transfer of energy, mass, and momentum across the ocean-atmosphere boundary. Let us begin by exploring how these transfers work.

To begin with energy, an important difference between the ocean and the atmosphere is how they interact with solar radiation. The atmosphere is largely transparent to solar radiation, but the ocean strongly absorbs it. The ocean surface reflects some of the solar radiation

reaching it, but most is absorbed and thus heats the upper 10 or so meters (33 feet) of the ocean.

The warm ocean-surface layer is a fundamental player in maintaining Earth's energy balance by returning energy to the atmosphere. It does so by various means. By far the most important is latent-heat transfer, or the exchange of heat when matter changes from one phase to another, as when liquid water evaporates to form water vapor (chapter 2). About $4.49 \times 10^{14} \text{ m}^3$ of water evaporates from the ocean each year, an amount almost 20 times that contained in the Great Lakes!¹ Evaporating that much water takes a tremendous amount of energy. Assuming an average surface density of the ocean (1.027 g/cm^3) and a latent heat of vaporization for water of $2,260 \text{ J/g}$, the amount of energy required annually is about $1 \times 10^{24} \text{ J}$, or about 2,000 times the contemporary annual global human energy consumption. This energy removed from the ocean by evaporation warms the atmosphere when the water vapor eventually condenses back into liquid water (to form clouds or precipitation).²

The latent-heat transfers occur unevenly across the ocean-atmosphere interface because of spatial variations in incident sunlight, cloudiness, windiness, atmospheric humidity, and ocean currents, to name a few important factors. The latent heat is also released unevenly vertically in the atmosphere, depending locally on the amount of water that is evaporated, the air temperature, the vertical structure of the atmosphere, and the strength of convection. These uneven distributions ultimately result in differential heating of the atmosphere and represent one mechanism that produces further variability in the state of the ocean and atmosphere.

Sensible-heat transfer is another means by which heat moves back and forth between the ocean and the atmosphere. Sensible heat (chapter 2) refers to the conductive transfer of heat from a warmer body in contact with a cooler body, such as from warmer ocean to cooler air. Importantly, when air is warmed by the ocean, it convectively rises and is replaced by cool air, which, in turn, is available to be warmed by the ocean. This makes the process much more efficient than it otherwise would be. Thus, although sensible-heat transfer is not nearly as important as latent-heat transfer, it can be relatively important in regions where strong ocean currents bring heat from equatorial regions to the high latitudes. For instance, the Gulf Stream moves a tremendous amount of heat from the tropical Atlantic Ocean and the Gulf of Mexico to the cold North Atlantic (chapter 2). There, the ocean transfers large amounts of sensible heat to the atmosphere, particularly during the winter, when the atmosphere is much colder than the underlying ocean. Conversely, if overlying air is warmer than the ocean, heat conductively flows into the ocean, but this process is relatively inefficient because as the air cools, the atmosphere becomes more stably stratified and cannot bring more heat to the interface.

The transfers of mass between the ocean and the atmosphere are, in addition to the transfers of energy, fundamental in determining the characteristics of the climate system. The water evaporated from the ocean results in the formation of clouds, in precipitation, and in humidity variations throughout the global atmosphere. Moisture is, of course, returned to the ocean through precipitation, either directly or through runoff from land. Evaporation also drives global variations in ocean salinity, which, in turn, help drive thermohaline circulation (chapter 2). The uptake or release of carbon dioxide (CO_2) by the ocean from or to the atmosphere is the primary natural control on the CO_2 content of the atmosphere and

thus on global temperature over timescales of decades to centuries and longer (chapter 4). The atmosphere also carries dust from the continents to the oceans that fertilizes the oceans with minerals that are critical for many marine organisms.

As for the transfer of momentum, winds impart a stress on the ocean surface through friction, largely driving ocean-surface currents and accounting for patterns of circulation embodied in the basin-scale ocean gyres (chapter 2). The interaction of the winds, ocean currents, and radiative heating can impose important regional circulation patterns in both the atmosphere and the ocean. Ocean-surface temperatures, for instance, can alter regional pressure patterns in the atmosphere and thus cause changes in winds, move convection regions, and alter the direction of storm tracks and jet streams. Winds, in turn, feed back on the ocean by changing the direction of ocean currents and influencing the transfer of energy between the atmosphere and the ocean.

These examples of energy, mass, and momentum exchanges together serve to illustrate how intimately the ocean and the atmosphere are connected through physical and chemical interactions (Back-of-the-Envelope Calculation). These interactions are fundamental for driving processes in the climate that generate a broad and complex spectrum of variability that remains at the center of our efforts to understand and characterize the system.

The El Niño–Southern Oscillation

Every several years, a dramatic and rapid warming occurs in the tropical Pacific Ocean off the coast of Peru and Ecuador. Within a month, water temperature rises by 2 to 4°C (4–7°F). Anchovies disappear; sardines move south; and birds, fur seals, and other animals whose livelihoods depend on the fish die. Heavy rains and flooding inundate the normally arid coastal regions of northern Peru and Ecuador, while droughts strike the Andes of southern Peru and more distant regions such as eastern Australia and northeastern Brazil. The peak warming of the tropical Pacific usually takes place around Christmas and persists into May or June. Local fishermen have long referred to this periodic occurrence as *El Niño*, the Spanish phrase for the “Christ Child,” because of its timing and the perceptible impacts that it has on coastal fisheries when it occurs.

El Niño and its counterpart, *La Niña*, are ocean phenomena, but they are intimately connected to the atmospheric phenomenon known as the Southern Oscillation, which is a swing in atmospheric pressure between the eastern and the western parts of the equatorial Pacific Ocean. Because the two are coupled, they are referred to collectively as the *El Niño–Southern Oscillation* (ENSO). While the processes that drive ENSO are specific to the tropical Pacific Ocean and surrounding regions, the impacts of ENSO are felt globally. *El Niño* commonly creates severe drought in Australia, Indonesia, southern Africa, and the African Sahel (the southern borderland of the Sahara Desert stretching from Senegal in the west to Sudan in the east); weakens the South Asian monsoon; and brings increased winter precipitation to parts of North America. It even affects the formation of hurricanes in the North Atlantic, where these storm systems are markedly less common during *El Niño* than in other years.³ *La Niña* similarly has significant global impacts. It causes droughts across the western United States and Mexico, brings heavy rains to northwestern Australia and Indonesia during the



BACK-OF-THE-ENVELOPE CALCULATION

Storing Heat in the Atmosphere and Ocean

The higher heat capacity of the oceans, relative to the continents, is important on daily and seasonal timescales because the land surface heats up and cools off much faster than the oceans. The heat capacities of the different components of the climate system are also important in the storage of energy gained as a consequence of human-caused global warming. As climate has warmed, different amounts of energy have been absorbed by the oceans, continents, atmosphere, and cryosphere.

Because of its large heat capacity, mass, and broad spatial coverage, the uppermost 700 meters (2,297 feet) of the global ocean has absorbed the most heat over the past several decades: 173×10^{21} J from 1971 to 2010. The continents and atmosphere, in contrast, have absorbed considerably less heat, in part because of their smaller heat capacities: from 1950 to 2000, the continents absorbed 9.7×10^{21} J, and from 1979 to 2010, the atmosphere absorbed 2.0×10^{21} J.¹

It is illustrative to consider the approximate temperature change associated with ocean heat gain and how much the atmosphere would have warmed if it had absorbed the same amount of heat as the upper ocean. To make those estimates, we use the formula for heat capacity that relates the amount of heat added (ΔQ) to the temperature change (ΔT) of a given substance,

$$\Delta Q = mc_p \Delta T,$$

where m is mass and c_p is heat capacity.

Let's use this equation to first derive an estimate for the temperature change of the upper ocean based on the estimate of heat gained. The heat capacity of ocean water is approximately 4,000 J/kg K. To determine the mass of the upper ocean, let us assume that ocean water has a density of 1,000 kg/m³ (while realizing that the density of ocean water varies with temperature and salinity). This density can be converted to mass by estimating the volume. The ocean covers roughly 71 percent of Earth's surface (5.10×10^{14} m²), yielding approximately 3.62×10^{14} m² of ocean-surface area. The volume of water is therefore the depth (700 meters) multiplied by the surface area and equal to 2.53×10^{17} m³ (here we neglect the sphericity of Earth). If we now use the density and heat capacity of ocean water, and the volume of the first 700 meters of the world ocean, we can solve for ΔT in $\Delta Q = mc_p \Delta T$ to estimate an average temperature change of 0.17°C (0.31°F).

But what about the troposphere? It gained much less heat than the ocean from 1979 to 2010, and yet it has warmed a similar amount (~ 0.44 – 0.54°C [0.79 – 0.97°F] in the lower troposphere and ~ 0.14 – 0.41°C [0.25 – 0.74°F] in the mid-troposphere from 1979 to 2012). Part of this is explained by the different heat capacities and masses of the ocean and the atmosphere. To illustrate, let's consider how much the troposphere would have warmed if it had absorbed all the heat that went into the upper ocean from 1971 to 2010. To make this approximation, we use $\Delta Q = mc_p \Delta T$, but estimates for the heat capacity of air (1,000 J/kg K) and the mass of the troposphere. For the latter, we will use the fact that it comprises about 75 percent of the total atmospheric mass (5.15×10^{18} kg).² Given these numbers, we can solve for ΔT and estimate that the troposphere would have warmed a whopping 45°C (81°F) if all the heat stored in the

upper ocean between 1971 and 2010 had instead gone into the troposphere! That amount of warming would make for a different world indeed.

What assumptions have we made that affect our calculations? First of all, it is important to note that there are many reasons why the atmosphere and the ocean absorbed their respective amounts of heat over the latter part of the twentieth century and into the twenty-first. While our calculations have been illustrative, the troposphere would never have absorbed all the heat that went into the ocean given the imposed conditions. It is also important to note that each of the numbers we used is subject to uncertainties, and although we have treated them as single values, they all are reported with uncertainty ranges that we have not included. We have also estimated the specific-heat capacities and mass of ocean water and air as single quantities, when in fact they fluctuate as functions of temperature, salinity (ocean), and water vapor (air). We have also treated both the ocean and the atmosphere as homogeneous volumes in which heat is stored evenly. This is decidedly not the case and will affect how heat is stored and temperatures change vertically and horizontally in each medium. For instance, we estimated that the average temperature change of the first 700 meters of the ocean was 0.17°C (0.31°F). This is comparable to measurement-based estimates of the temperature change in this layer, but the warming from 1971 to 2010 was larger over the first 75 meters (246 feet) of the ocean ($\sim 0.44^{\circ}\text{C}$ [0.79°F]) and diminished at 700 meters ($\sim 0.06^{\circ}\text{C}$ [0.11°F]). As noted, similar measurements in the atmosphere show different warming trends throughout the troposphere over the same time period.

Our back-of-the-envelope calculation nevertheless has demonstrated the importance of heat capacity and the masses of the ocean and atmosphere in governing their temperature changes given equivalent heat gains.

1. Heat gain estimates are derived over different intervals based on the availability and characteristics of the data. All values are taken from D. L. Hartmann, "Observations: Atmosphere and Surface," in *Climate Change 2013: The Physical Science Basis. Contribution of Working Group I to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change*, ed. T. F. Stocker et al. (Cambridge: Cambridge University Press, 2013), 159–254.

2. Atmospheric mass is estimated in K. E. Trenberth and L. Smith, "The Mass of the Atmosphere: A Constraint on Global Analyses," *Journal of Climate* 18 (2005): 864–875.

boreal (Northern Hemisphere) winter, and enhances the South Asian monsoon during the boreal summer. These events have important implications for countries and communities that inhabit these regions, making ENSO an important system to study.

ENSO DYNAMICS

El Niño and La Niña are consequences of interactions between the atmosphere and the ocean across the equatorial Pacific Ocean and represent a dramatic example of ocean and atmosphere coupling. El Niño occurs at irregular intervals of two to seven years and has been a feature of the climate system for at least the past 130,000 years. Next to the seasons, it is the most important recurrent change in Earth's climate.

How does ENSO work? In order to understand the coupled dynamics that create ENSO, it is necessary to first understand the tropical Pacific Ocean in its so-called neutral state. In this state, the easterly trade winds of the Northern and Southern Hemispheres converge at the equator and blow west across the equatorial Pacific. The wind stress produces a corresponding westward-flowing equatorial current (figure 3.1). This current causes the equatorial surface waters, which are warmed by the strong radiative heating from sunlight through the tropics, to accumulate in the western Pacific and form the *Western Warm Pool* (WWP [see figure 3.1]). Indeed, so much water piles up that the sea level of the western tropical Pacific is typically about 60 to 70 centimeters (23–27 inches) higher than that of the eastern tropical Pacific (figure 3.2).

The easterly winds cause Ekman transport (chapter 2) in the tropical ocean, carrying water away from the equator and causing upwelling of cold water in a narrow band along it. Additionally, the strong coastal Peru Current causes upwelling along the west coast of South America. The upwelling brings the thermocline (along with nutrients that are the basis of the fertile fisheries in the region) to the surface in the east, thus making the surface waters of the eastern equatorial Pacific much colder than those of the western Pacific. As the surface waters flow westward, a strong equatorial “undercurrent,” typically at a depth of about 20 to 400 meters (65–1,300 feet), flows eastward to replace the upwelling in the east. In the meantime, the accumulation of warm waters in the west forces down the thermocline beneath the WWP to a typical depth of about 150 meters (490 feet).

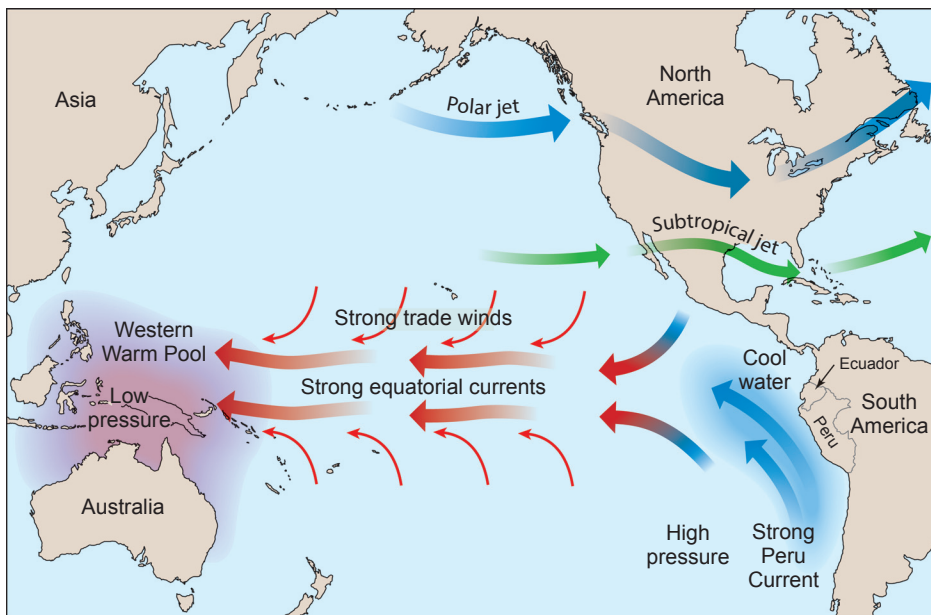


FIGURE 3.1 The neutral state of the tropical Pacific Ocean

The convergence of the trade winds along the equator causes wind stress on the ocean surface and a surface current with a mean flow from east to west. The current causes warm water to collect in the western tropical Pacific and establishes the Western Warm Pool. Upwelling and Ekman divergence cause colder sea-surface temperatures in the east.

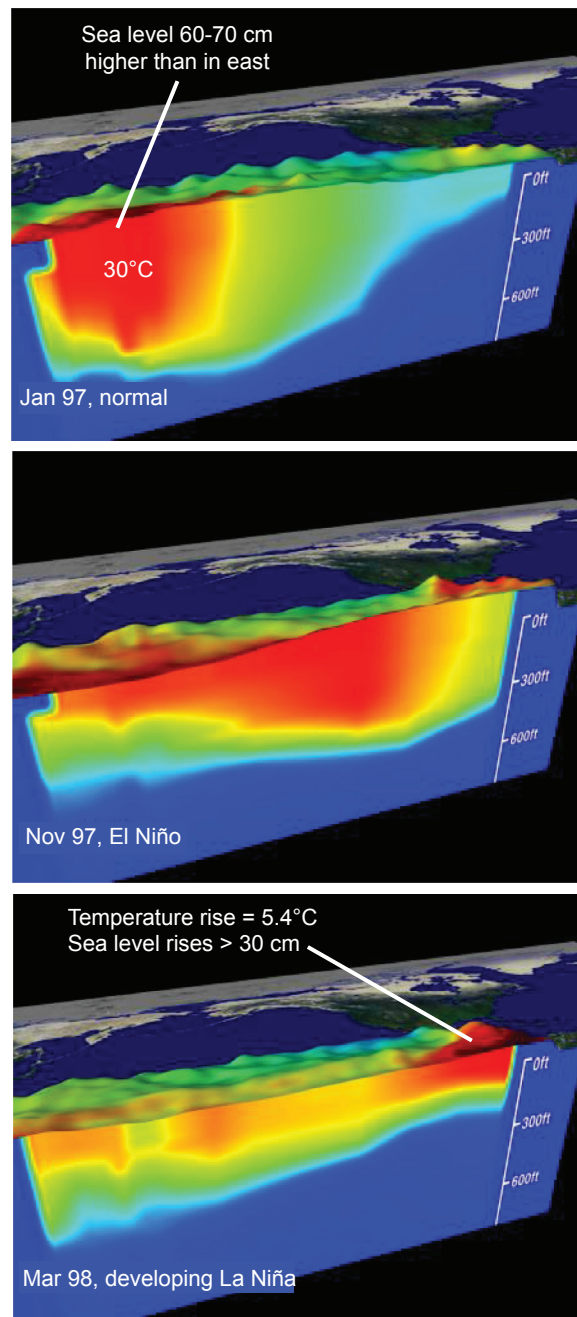


FIGURE 3.2 Changes in ocean temperature and sea-level height in the equatorial Pacific Ocean during the 1997–1998 El Niño

Before the development of El Niño conditions in January 1997, the ocean temperature and sea-level height in the tropical Pacific reflected neutral state conditions. The thermocline depth was deeper in the west, reflected by warm, 30°C (86°F) temperatures extending to more than 150 meters (490 feet) beneath the WWP. In the east, the thermocline was much shallower, with colder, 8°C (46°F) temperatures extending almost to the surface. Once the El Niño developed in late 1997 and into 1998, sea level flattened and warm temperatures extended across the whole of the equatorial region. The thermocline therefore shoaled in the west and deepened in the east, where sea surface temperatures increased and warm waters extended to depths of more than 100 meters (330 feet). (NOAA, Pacific Marine Environmental Laboratory, "Sea Level & Temperature," El Niño Theme Page, <https://www.pmel.noaa.gov/el-nino/el-nino-and-sea-level>)

In the atmosphere, a pressure contrast develops between the eastern and the western parts of the equatorial Pacific Ocean. In the west, the warm ocean and land surfaces heat the atmosphere and create a low-pressure center over the WWP, while in the east, the colder water creates a high-pressure center. The east–west pressure contrast drives easterly surface winds along the equator, thus enhancing the trade winds that began this discussion. The warm, moist convectively rising air over the WWP brings precipitation to that region. The rising air also feeds a returning, eastward-directed flow of air near the top of the troposphere that eventually sinks in the eastern Pacific, feeding the zone of high surface pressure in that region (figure 3.3). This equatorial convective cycling of the atmosphere is known as *Walker circulation*, for the British physicist and statistician Sir Gilbert Walker (1868–1958), who, studying records of the South Asian monsoon while stationed in India, first recognized the relationship between the strength of the monsoon and changing wind patterns in the Pacific Ocean (Historical Note).

It is worth pausing here to emphasize the strong positive feedback between the ocean and the atmosphere in the coupled neutral state just described. The trade winds, ultimately the product of convergence at the equator driven by strong equatorial insolation (chapter 1),

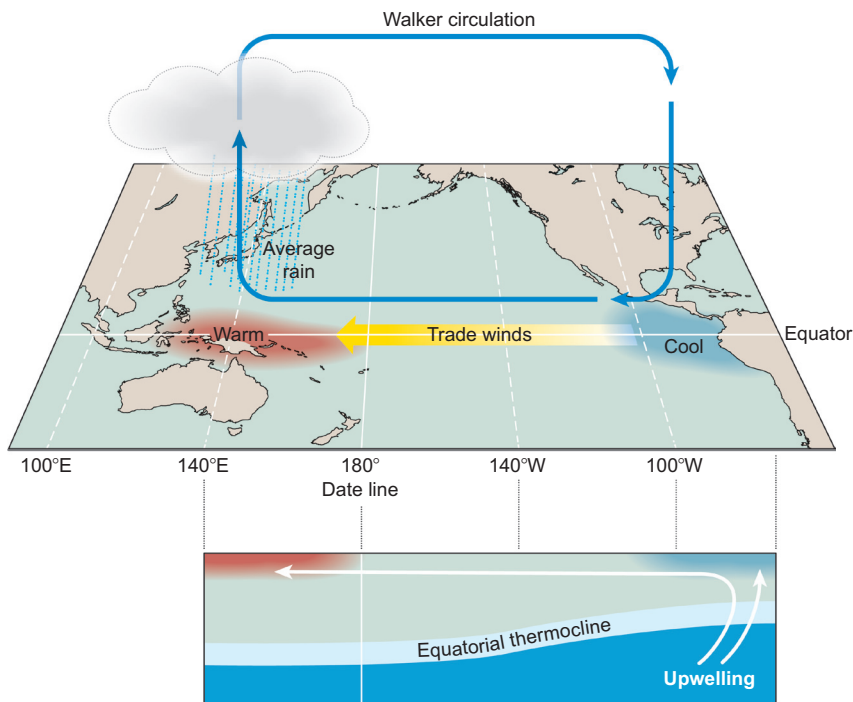


FIGURE 3.3 Neutral conditions over the equatorial Pacific Ocean

During neutral conditions, the WWP causes convection in the western Pacific, which feeds precipitation. Aloft, the rising air diverges eastward and then descends in the eastern equatorial Pacific, completing the Walker circulation cell. The contrast between the high surface pressure in the east and the low surface pressure in the west produces strong easterly winds along the equator. In the ocean, the thermocline deepens underneath the WWP, while it shoals in the east. (Redrawn from Australian Government, Bureau of Meteorology, “The Three Phases of the El Niño–Southern Oscillation [ENSO],” <http://www.bom.gov.au/climate/enso/history/ln-2010-12/three-phases-of-ENSO.shtml>)



HISTORICAL NOTE

Sir Gilbert Walker, Monsoons, and the Southern Oscillation

Gilbert Walker (1868–1958) was raised in Croydon, now a part of Greater London, but when he was growing up, it was a leafy suburb about 15 kilometers (9 miles) south of the city center.¹ He was the fourth of eight children born to Elizabeth Haslehurst, who became an invalid when Gilbert was quite young, and Thomas Walker, the borough engineer of Croydon, who was known for his pioneering use of concrete for large structures such as dams. After attending the local grammar school, Gilbert obtained a scholarship to study at St. Paul's School, a selective public school in West Kensington, London: "There, according to his own account, he spent two years on the classical side but was then 'sent in disgrace' to the mathematical side when he made some appalling linguistic blunder."² Apparently, the change was not so bad because he discovered interests in both mechanics and mathematics. For example, he received a prize, at age 17, for a gyroscope that he had built, and his proclivity for mathematics earned him a scholarship to study the subject at Trinity College, Cambridge, where he began as an undergraduate in 1886.

At Trinity, Walker established himself as one of the top students, winning awards and becoming "first wrangler" (gaining first place) in the 1889 Mathematical Tripos I, Cambridge's notably difficult and culminating examination needed for the B.A.³ He then continued for another year to the Mathematical Tripos II, where he also emerged with the leading grade and a degree in mathematics. Along the way, he became fascinated with, of all things, boomerangs, apparently introduced to them during a trip to Australia. After graduation, he spent three winters in Switzerland recovering from health problems, and there he developed a passion for ice-skating. In 1895, he became a lecturer at Trinity and wrote several papers on mathematics, among the most notable of which was one on the aerodynamic forces acting on the boomerang. That paper led to another concerning the mechanics governing the motion of objects in a variety of sports such as billiards, golf, and cycling.

While Walker was thinking about boomerangs, halfway around the world, Sir John Eliot (1839–1908), an English meteorologist and the second director of the India Meteorological Department (IMD or Met Department), was struggling to find ways to accurately forecast the monsoonal rains. This was of major concern to the people of India because failures of the monsoon commonly resulted in famine. The IMD, in fact, had been founded in 1875 to better forecast the monsoon after devastating famines in 1866 and 1871. While the IMD had developed expertise in daily forecasting, it was unable to make significant progress in seasonal forecasts. Eliot realized that long-term prediction would likely rest on finding statistical correlations between monsoons and various weather data—in other words, prediction was partly a mathematical exercise. Advice from the Observatories Committee of the Royal Society led Eliot to Walker. Apparently, this is how a mathematician with no training whatsoever in meteorology became the director of the IMD.

Walker departed Trinity College for India in 1903 and took over the directorship of the IMD the following year. The choice, if unconventional, turned out to be brilliant:

When Walker started work he realized at once that there was little scientific basis for the production of seasonal forecasts. His whole previous scientific career had been devoted to problems in which results could be produced by mathematical analysis from an assured

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basis of simple principles. It is a sign of the high quality and flexibility of his mind that he could realize that all the skill he had acquired in the past would be of little use to him in his new situation. He decided that since he saw no prospect of treating the weather as a subject to which mathematical reasoning from well established premises could be applied, he would collect all the relevant information which had been recorded and treat it statistically without attempting to trace physical connexions between cause and effect.⁴

Two elements of the investigation would become important. First, recognizing that the monsoon is a regional, not a local, phenomenon, Walker came to appreciate the need to search for statistical correlations with distant meteorological phenomena. This was an enormous task; initially, he had well-trained scientific assistants to conduct the search, but during World War I and later, he established a “human computer” made up of a number of clerical and other Indian staff to search for the correlations. Second, he sought “lag correlations,” or those for which there are time delays, leading to rigorous methodologies to treat time series and to test the apparent statistical correlations. Thus was born the concept of “world weather,” leading to a series of memoirs later in his career, all of which had the main title *Correlation in Seasonal Variations of Weather*. The approach led him to the discovery of the oscillation in air pressure over the Indian and Pacific oceans (the latter he termed the Southern Oscillation) and their correlation with rainfall in various parts of the world. Walker recognized, however, that while the air-pressure variations can influence the strength and local distribution of the monsoon, the statistical relationship between Indian Ocean and Pacific Ocean pressures was not a forecast. Rather, the statistical relationship provides what he would call a “seasonal foreshadowing” of events to come. Farther afield, the approach pioneered by Walker also led him to the discovery of the North Atlantic Oscillation.

Walker’s life in India was full. In 1908, he married Mary Constance Carter, with whom he had two children. During the summers, he worked and the family stayed in Simla (Shimla), in northern India at an elevation of about 2,200 meters (7,200 feet) and then the summer capital of the British. There he pursued his many interests. In addition to throwing boomerangs, he became fascinated by the aerodynamics of bird flight, about which he wrote several papers; played the flute and conducted scientific studies of the instrument itself, resulting in a new instrument design; painted with watercolors; and ice-skated (a popular sport in northern India in towns with outdoor rinks).

Walker returned to England in 1924 and was soon knighted. He was appointed professor of mathematics at Imperial College, London, and retired, in 1934, to Cambridge, where he continued to write, enjoy considerable recognition, and immerse himself in meteorology. During World War II, he served as an honorary consultant in long-range weather forecasting, a new activity for the military that would prove to be important for the war effort. His last paper was published in 1947. He remained in Cambridge for several more years and then apparently led a more itinerant life with family between mainly Sussex and Surrey, where he died on November 4, 1958, at age 90.

1. This historical note is based primarily on G. I. Taylor, “Gilbert Thomas Walker, 1868–1958,” *Biographical Memoirs of Fellows of the Royal Society* 8 (1962): 166–174. Other sources include R. Allan, “Gilbert Walker: A Pioneer of Modern Day Climatology,” Walker Institute. <http://www.walker-institute.ac.uk/media/GILBERT>

_WALKER.pdf; and J. M. Walker, “Pen Portrait of Sir Gilbert Walker, CSI, MA, ScD, FRS,” *Weather* 52 (1997): 217–220.

2. Taylor, “Gilbert Thomas Walker,” 167.

3. For the etymology of the word *wrangler* and a “Guide to the Mathematical Tripos,” see University of Cambridge, Faculty of Mathematics, “Undergraduate Mathematics: Course Information,” <https://www.maths.cam.ac.uk/undergrad/course>.

4. Taylor, “Gilbert Thomas Walker,” 170.

establish the east–west contrast in sea-surface temperatures (SSTs) across the equatorial Pacific Ocean. The SSTs, in turn, establish regions of low atmospheric pressure in the west and high atmospheric pressure in the east, a contrast that enhances the easterly winds and further maintains the SST pattern. Strong convection over the WWP feeds one arm of the Walker circulation, which, in turn, feeds descent and the high surface pressures over the eastern Pacific. The strong westward-flowing equatorial current causes warm water to build up in the WWP and deepens the thermocline there, while wind-driven Ekman transport and upwelling feed the cooler SSTs in the east. Because these are all positive feedbacks, one might imagine them working indefinitely to maintain the neutral state in the equatorial Pacific Ocean, but it also means that these feedbacks are subject to collapse when perturbations occur.

What happens if there are changes to the neutral conditions in the equatorial Pacific Ocean? For instance, what if the trade winds diminish in strength? When such a change occurs, the strength of the surface currents in the tropical Pacific weaken. This, in turn, slows the transport of warm waters from east to west and reduces the SST gradient along the equator. These changes then decrease the east–west atmospheric-pressure difference, diminishing the pressure gradient across the equatorial Pacific, reducing the easterly winds across the equator, and further weakening the surface current that creates the WWP and causes Ekman divergence and upwelling in the east. As these feedbacks grow, the neutral state of the system collapses, and El Niño conditions develop.

During an El Niño year, the easterly trade winds weaken, warm western Pacific Ocean water flows back toward the east, and sea level flattens out (figure 3.4; see figure 3.2)—in essence, the water that had piled up in the WWP sloshes back to the east in an enormous (but imperceptible to us) wave. Upwelling ceases along the coast of South America and along the equator, the thermocline deepens to tens of meters in the east, and the equatorial undercurrent stops. The warmer water moving east now heats the air over the central and eastern Pacific, causing convection over those regions and bringing heavy rains as far east as coastal South America. In the meantime, because the WWP is diminished and the low pressure has moved to the east, dry air commonly descends over Indonesia and Australia to create drought conditions there. Overall, these changes contribute to a weakened Walker circulation.

While El Niño represents a breakdown of the neutral conditions in the tropical Pacific Ocean, La Niña represents the opposite—the intensification of the neutral state (see figure 3.4). La Niña conditions occur when the easterly winds along the equator strengthen, driving the WWP farther west and causing SSTs in the region to rise. Convection over the

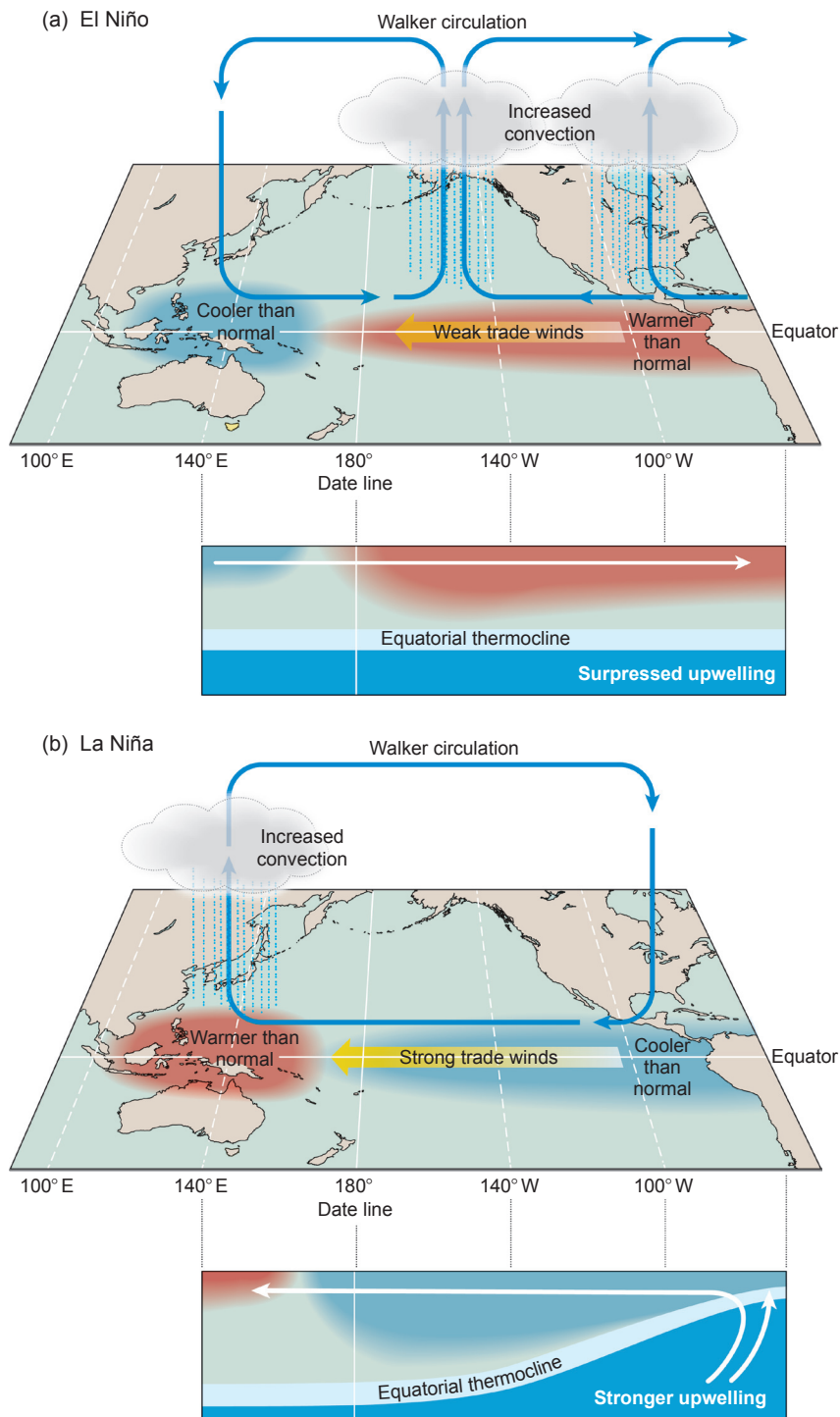


FIGURE 3.4 Tropical Pacific conditions characteristic of El Niño and La Niña

During El Niños (a), the trade winds collapse, and water in the WWP flows back to the east. The atmospheric convection center moves to the central and eastern Pacific Ocean, and Walker circulation weakens. In the ocean, the thermocline shoals in the west and deepens in the east. During La Niñas (b), the neutral state is intensified. The trade winds strengthen and drive the WWP and center of atmospheric convection even farther to the west, while the thermocline deepens in the west and shoals even more in the east. The Walker circulation also intensifies. (Redrawn from Australian Government, Bureau of Meteorology, "The Three Phases of the El Niño–Southern Oscillation [ENSO]")

WWP also moves west and intensifies, bringing more precipitation to parts of Australia and Indonesia. In the meantime, the stronger easterly equatorial winds enhance upwelling in the eastern Pacific, reducing ocean-surface temperature. The colder waters increase the atmospheric high pressure, which acts to suppress precipitation over the eastern Pacific. The higher atmospheric pressure in the east and the lower pressure in the west increase both the surface and the high-altitude east–west pressure gradients and thus strengthen the Walker circulation.

ENSO VARIABILITY

Measuring ENSO is a fundamental part of how it is observed and how its variability is characterized. One means of measuring the system is through atmospheric pressure. The Southern Oscillation Index (SOI) is a long-standing measure of the atmospheric-pressure difference between Tahiti and Darwin, Australia, reflecting the pressure difference over the WWP and the far westward extent of the cold eastern equatorial waters. Analogous to the SOI, ocean ENSO indexes are constructed from averages of SST anomalies over geographic boxes centered over different regions of the tropical Pacific Ocean.⁴

Let's consider the Niño3.4 index (figure 3.5), which is the average SST in a box within the tropical Pacific extending from 5°N to 5°S and from 170°W to 120°W. We present the monthly index back to 1900 to illustrate the long-term variability of ENSO. The record first indicates that El Niño occurs on average about every four years. In that sense, the events are quasi-periodic, but at the same time their occurrence is irregular. For example, between the El Niño years of 1977–1978 and 1986–1987, there was one El Niño (a very large one!) in 1982–1983, marking three-year intervals between each of the other two events. In contrast, from

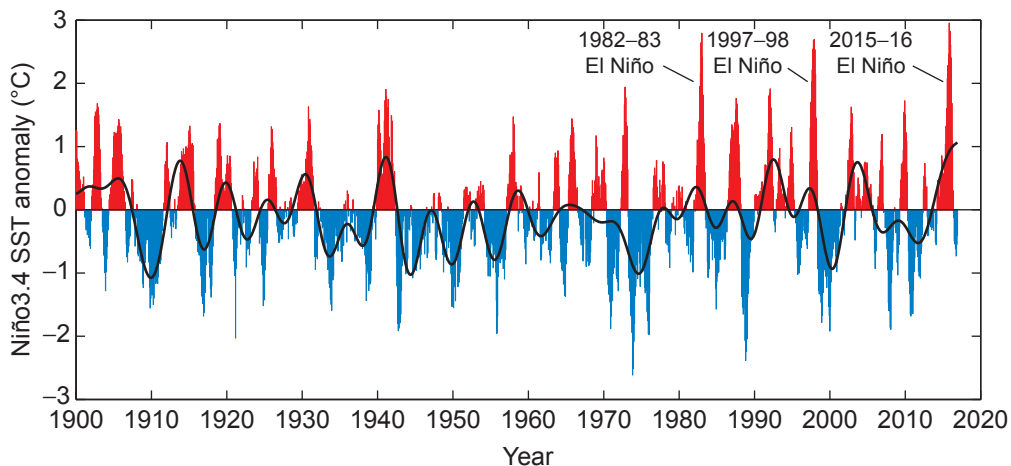


FIGURE 3.5 The monthly Niño3.4 index, 1900–2016

The index represents the average sea-surface temperatures in a geographic box covering the region bounded by 5°N–5°S latitudes and 170°W–120°W longitudes. The index is negative during La Niña and positive during El Niño. The thicker black line is a filtered version of the index that highlights the ENSO variability at five years and longer. (Data from KMNI Climate Explorer, “Niño3.4 Index,” <https://climexp.knmi.nl/data/inino5.dat>)

1991 to 1998, several El Niños followed each other in close succession, with only a few modestly cold periods throughout the interval. Similar periods of quiescence or enhanced activity exist for La Niñas. Second, some ENSO events are intense, while others are not. The El Niños of 1982–1983, 1997–1998, and 2015–2016 were three of the most intense of the past century. Other El Niños were barely noticeable, however, and existed very close to the neutral state. Third, El Niños and La Niñas alternate back and forth in a quasi-periodic fashion with an average timescale of two to seven years. In fact, El Niños and La Niñas can simply be viewed as two alternating states of ocean–atmosphere conditions, with neither one nor the other being abnormal.

So what causes ENSO variability from year to year? This question is still actively debated. One theory holds that oscillatory processes that are a fundamental part of the ENSO system are responsible, while another asserts that the oscillations are maintained by external and random weather influences.⁵ Both theories have been successful at explaining aspects of the ENSO system, but their applicability is still debated. Nevertheless, pursuing these theoretical characterizations is not just academic because they underlie our ability to predict ENSO events before they happen.

Regardless of what actually drives ENSO variability from year to year, it is clear that seasonal oscillations in the atmosphere and random variations in the strength of the trade winds and convection over the WWP can cause the winds to strengthen or weaken enough to initiate the feedbacks that develop into an El Niño or a La Niña. In the ocean, waves that travel back and forth across the tropical Pacific basin shift the depth of the thermocline and therefore influence the temperature of the upwelling waters in the east enough to do the same. These processes collectively interact to establish the rich range of variability in the ENSO system that is indicated by the Niño3.4 index in figure 3.5. But that is a picture of ENSO variability over roughly only the past 150 years. Paleoclimatic evidence⁶ and climate models indicate that ENSO exhibited an even wider range of variability in the more distant past. The natural variability of the ENSO system is therefore reasonably well documented, but a pertinent question today is the extent to which ENSO will respond to global warming. Suffice it to say here that ENSO does appear to respond to warming, but how and by how much are uncertain.

ENSO IMPACTS

El Niño and La Niña affect weather conditions around the globe. The most direct impacts are felt in the tropical Pacific Ocean, where the dynamics of ENSO cause shifting convection and rainfall patterns. Recall that during the neutral state, the WWP and regions of Indonesia and Australia experience wetter conditions, relative to the drier and colder regions in the eastern equatorial Pacific and over Ecuador and Peru. During La Niña events, the neutral conditions are intensified and can lead to heavy precipitation and flooding in areas in and around the WWP, while dry conditions characterize the eastern Pacific. During El Niño events, the convection shifts away from the WWP and into the central and eastern Pacific. This brings dry conditions to Indonesia and Australia and strong rainfall to the central and eastern Pacific.

The impacts of ENSO nevertheless are not confined to the tropical Pacific. ENSO, in addition to other coupled ocean–atmosphere phenomena, influences global weather patterns

through *teleconnections*, or shifts in the patterns of hemispheric and global climate that arise due to a regional phenomenon. During an El Niño, the strong convection center over the WWP moves eastward to the central and eastern Pacific. This change in location causes large-scale circulation patterns to change in the atmosphere. In particular, it shifts the jet streams to the south and amplifies mid-latitude storm systems (chapter 1). In North America during the boreal winter and early spring, these circulation changes generally bring cold and wet conditions to the southern United States and northern Mexico and warm and dry conditions to the Northwest, Central Plains, and parts of the eastern United States (figure 3.6). Farther afield, El Niño commonly brings dry conditions to the African Sahel, South Africa, and northeastern South America and disrupts the South Asian monsoon during the boreal

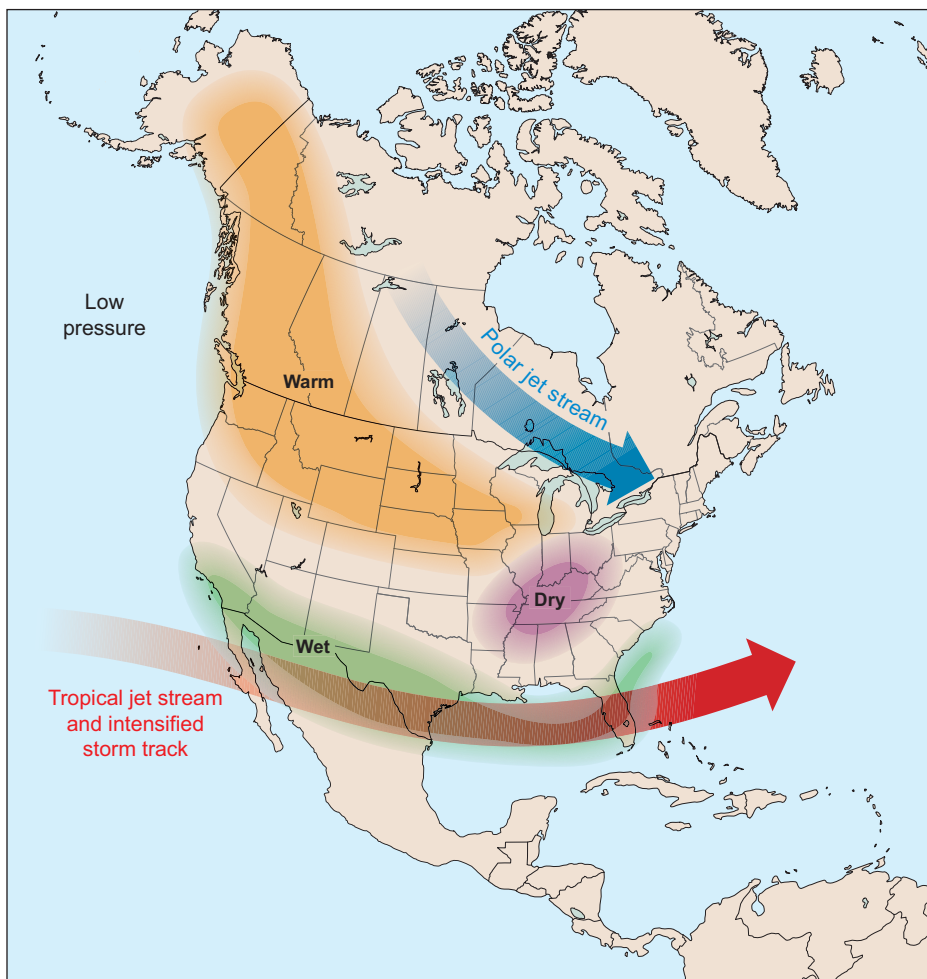


FIGURE 3.6 The common shifts in winter weather in North America during El Niño years

The change in weather is due in part to the southward shift of the subtropical jet stream and the intensification of storm tracks. This brings wet conditions to much of the southern United States, and warm and dry conditions to the northwestern and north-central states and western Canada. (Redrawn from map by F. Martin, in “El Niño and La Niña: Frequently Asked Questions,” January 18, 2016, <https://www.climate.gov/news-features/understanding-climate/el-ni%C3%B1o-and-la-ni%C3%B1a-frequently-asked-questions>)

summer by weakening the Walker circulation (figure 3.7*a*). La Niña also influences precipitation globally (see figure 3.7*b*). In North America, the jet streams and storm tracks shift north, causing dry conditions throughout the southern United States and northern Mexico. La Niña also tends to bring wet conditions to the African Sahel, southern Africa, and northern South America, while intensifying the South Asian monsoon.

Precipitation and temperature are not the only means by which ENSO influences weather. It also affects hurricanes in the Atlantic and Pacific Ocean basins. El Niño increases wind

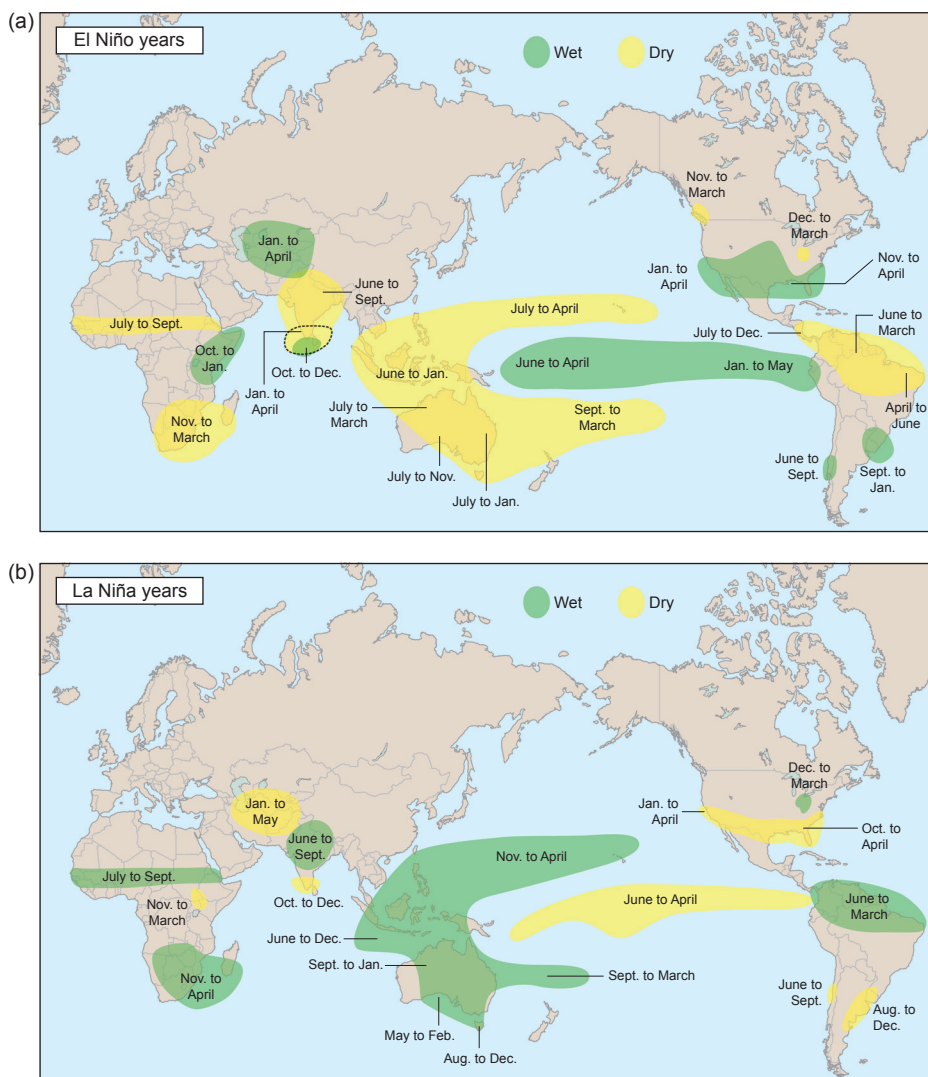


FIGURE 3.7 The typical shifts in global rainfall patterns associated with El Niño and La Niña

The shifts in rainfall are caused by teleconnections associated with the movement of the convection center in the tropical Pacific Ocean during (a) El Niño and with the westward movement of the WWP and intensification of the associated convection during (b) La Niña. (Redrawn from International Research Institute for Climate and Society, Earth Institute, Columbia University, “ENSO Essentials,” <http://iri.columbia.edu/our-expertise/climate/enso/enso-essentials/>)

shear (the shear caused by the differing speeds at which horizontal layers of the atmosphere move relative to each other) over the tropical Atlantic. Increased wind shear makes it more difficult for hurricanes to form, so this is one reason why fewer North Atlantic hurricanes occur during El Niño years than during others (figure 3.8). Conversely, because El Niño brings warmer SSTs across the whole of the tropical Pacific Ocean, hurricanes that form there during El Niño years tend to be more intense. ENSO also has recently been shown to affect tornado and hail activity in the tornado region of the United States.⁷ El Niño tends to suppress the occurrence of severe thunderstorms, which spawn tornadoes and hail, while La Niña enhances their occurrence.⁸

Through its influence on weather, ENSO also affects agriculture, ecosystems, infrastructure, water resources, and thus social systems. As noted in figure 3.7a, El Niño commonly brings drought to southern Africa. The effect on agriculture is demonstrated by the remarkable correlation between corn yield in Africa and SSTs in the eastern equatorial Pacific Ocean (figure 3.9). This general awareness of the connection between tropical Pacific conditions and agriculture goes back at least to the late 1890s, when Walker attempted to understand the connection among pressure oscillations in the western Pacific, the failure of the South Asian

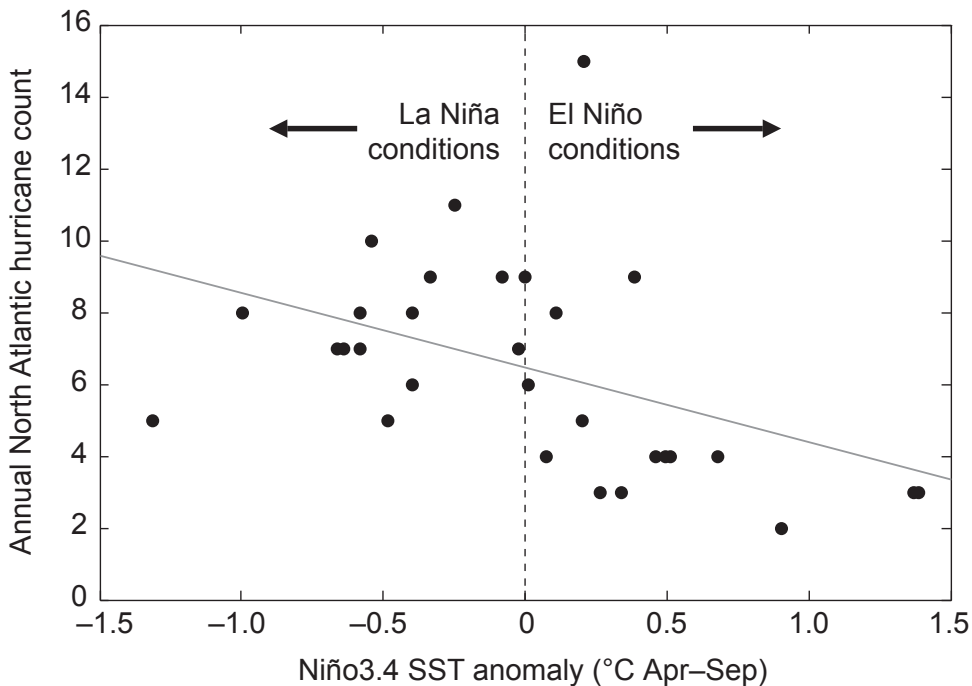


FIGURE 3.8 The relationship between average April–September sea-surface temperature in the Niño3.4 index and the annual hurricane count in the North Atlantic Ocean, 1979–2008

Fewer tropical storms and hurricanes occur over the North Atlantic sector during El Niño years and more during La Niña years. The gray line is the linear regression line between the two variables, which have a linear correlation of -0.45 . (Data from KMNI Climate Explorer, “Niño3.4 Index,” and “Adjusted Number of Hurricane Days Over the North Atlantic,” https://climexp.knmi.nl/data/ihurdat_hurday_adj.dat, the latter based on G. A. Vecchi and T. R. Knutson, “Estimating Annual Numbers of Atlantic Hurricanes Missing from the HURDAT Database [1878–1965] Using Ship Track Density,” *Journal of Climate* 24 [2011]: 1736–1746)

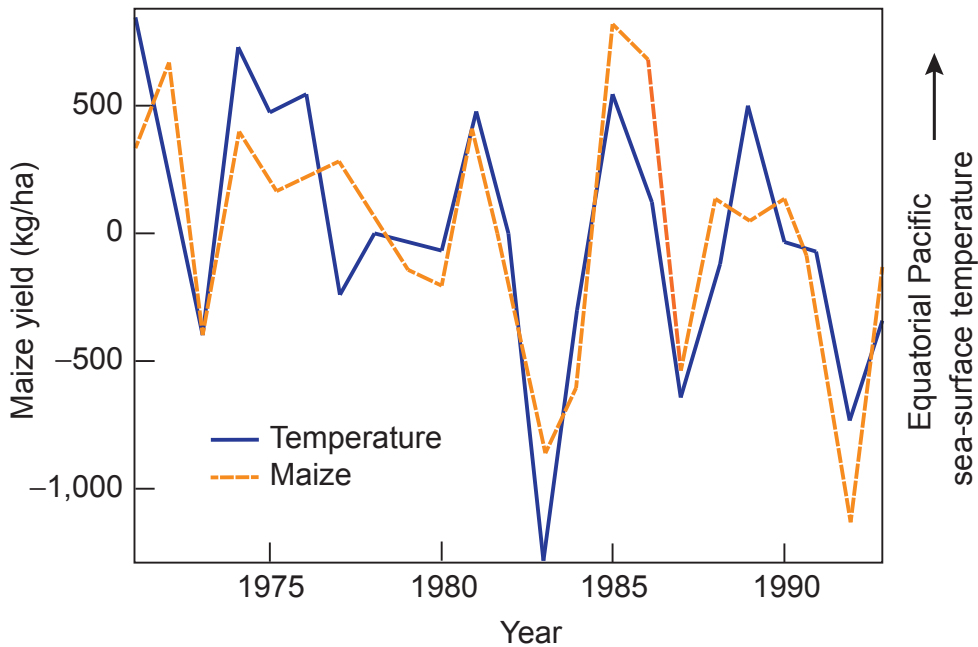


FIGURE 3.9 The relationship between annual maize yield in Zimbabwe and sea-surface temperature in the eastern equatorial Pacific Ocean

When these data were collected, the maize yield in Zimbabwe was a close reflection of rainfall. Sea-surface temperature is the average for February and March, the height of the growing season in Zimbabwe, for the region in the equatorial Pacific bounded by 90°W–150°W longitudes and 5°N–5°S latitudes (Niño 3). The correlation ($r=0.78$) demonstrates the effect that ENSO has on distant parts of the world. The maize yield is in kilograms per hectare relative to the national average. (Redrawn from M. A. Cane, G. Eshel, and R. W. Buckland, “Forecasting Zimbabwean Maize Yield Using Eastern Equatorial Pacific Sea Surface Temperature,” *Nature* 370 [1994]: 204–205)

monsoon, and the agricultural and social catastrophes that resulted. The effects of ENSO on society remain a focus of modern research, which has also connected ENSO variability to regional factors as specific as the incidence of snakebites in Costa Rica⁹ and as consequential as outbreaks of social conflict and violence.¹⁰

ENSO PREDICTIONS

Given the many impacts of ENSO variability, predicting the occurrence of El Niños and La Niñas has become an important goal of the scientific community, the idea being that with enough advanced warning, the people in regions that are commonly affected can make plans to blunt the consequences. Fortunately, the science has progressed so that useful predictions of ENSO events are now possible up to six to 12 months before they occur. This success starts with detailed observations of the tropical Pacific Ocean and atmosphere.

The Tropical Pacific Observing System is an international effort to collect data from satellites, ocean buoys, and radiosondes¹¹ to provide real-time measurements of wind, atmospheric pressure, SST, thermocline depth, and other relevant variables over a large rectangular block of the tropical Pacific Ocean. These measurements tell us what the state of the ENSO system currently is, but they are also used to predict what will happen in the future. This is

accomplished in much the same way as weather forecasting: the current observations are used to initialize mathematical models that calculate how the ocean and the atmosphere interact to exchange energy, mass, and momentum, as well as how the system evolves with time in response to those exchanges. Unlike weather models, however, the ENSO system is dependent on processes with longer timescales, so skillful predictions of ENSO can be performed many months in advance, as opposed to the five- to 10-day weather forecasts with which we are accustomed. Similar to weather forecasts, ENSO predictions are probabilistic; that is, they represent statistical characterizations of multiple prediction models. For instance, in March 2015, most models were predicting that an El Niño would develop later in the year, but several were not (figure 3.10). The composite forecast in March 2015 estimated a 50 percent probability of an El Niño developing from September to November 2015 (as opposed to a 10 percent probability of a La Niña developing then).¹² As the year progressed, the predicted probability of an El Niño increased, and a strong event did develop and intensify over the winter of 2015–2016. While scientists continue to improve their predictive models, the successful prediction in 2015 exemplifies the tremendous progress in our ability to characterize the ENSO system.

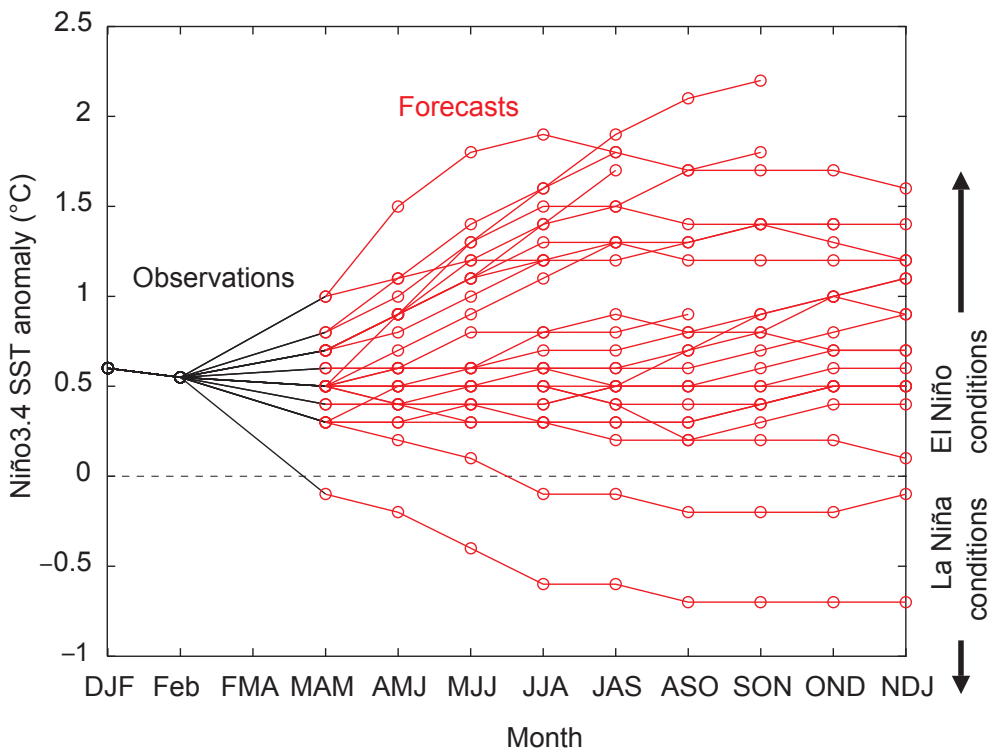


FIGURE 3.10 *Ensemble of El Niño predictions, March 2015*

The forecast models are initialized with observations of the tropical Pacific Ocean in March and then used to predict how the state of the ENSO will evolve multiple months into the future (each line represents a forecast from a different model). The forecast in March 2015 produced a 50 percent probability of an El Niño developing by September to November 2015 (as opposed to a 10 percent probability of a La Niña developing then). (Data from International Research Institute for Climate and Society, Earth Institute, Columbia University, “IRI ENSO Forecast: 2015 March Quick Look,” March 19, 2015)

Our success at short-term prediction raises the question of whether we can predict how ENSO will respond to continued global warming over the twenty-first century. Will it become more variable, changing from one state to the other more rapidly and frequently, or will the opposite occur? Can we expect more El Niños in a warming world, will La Niñas dominate, or will the ENSO system remain much like it did during the twentieth century?

There are a number of ways in which global warming might be expected to affect the ENSO system. If, for instance, global warming were to cause the central and eastern Pacific Ocean to warm more than the WWP, the east–west SST gradient would diminish. This would decrease the pressure difference between the east and the west, reduce the strength of the easterlies, slow the upwelling of cold water in the east, and lessen the strength of the Walker circulation—with the result that the neutral state of the tropical Pacific would look more El Niño-like. Alternatively, were the WWP to warm more than the eastern Pacific, the opposite string of feedbacks would occur and create mean conditions that are more persistently La Niña-like. Which one will it be? A clear picture has yet to emerge. Nevertheless, no one disputes that ENSO will continue its oscillations well into the future, regardless of how it changes as a result of global warming, and continue to significantly affect weather and connected systems around the globe. It is also important to note, however, that changes in ENSO cannot themselves be responsible for long-term increases or decreases in global temperatures like we have observed over the past century. ENSO simply influences how heat is partitioned between the ocean and the atmosphere, but it does not cause changes in the overall energy balance of the planet.

Other Modes of Ocean and Atmosphere Variability

While we have devoted most of this chapter to ENSO, there are, of course, other important oscillatory phenomena in the ocean and the atmosphere that occur on timescales similar to that of ENSO or over decades and longer. Some, particularly the monsoons, have direct and immediate bearings on the well-being of society.

MONSOONS

The term *monsoon* derives from the Arabic word *mawsim*, referring to the “season of winds,” and nowhere do people live more by a monsoon’s seasonal rhythm than in India, especially in the western part of the country, where the monsoon usually starts abruptly around mid-June and ends in September. The summer monsoon accounts for about 90 percent of the region’s yearly rainfall.

In the most simplistic of representations, the seasonal heating of the land and ocean plays the central role in establishing monsoons. Across much of South Asia, solar radiation during the boreal spring and summer heats the land faster than the surrounding ocean because of the differences in their heat capacities. This causes the convective rise of air over a zone of low atmospheric pressure on the land surface, which pulls in moist ocean air from the south and west, opposite the direction of the trade winds. This convection yields the high summer rainfall in the region (figure 3.11). The summer winds also drive ocean currents that flow

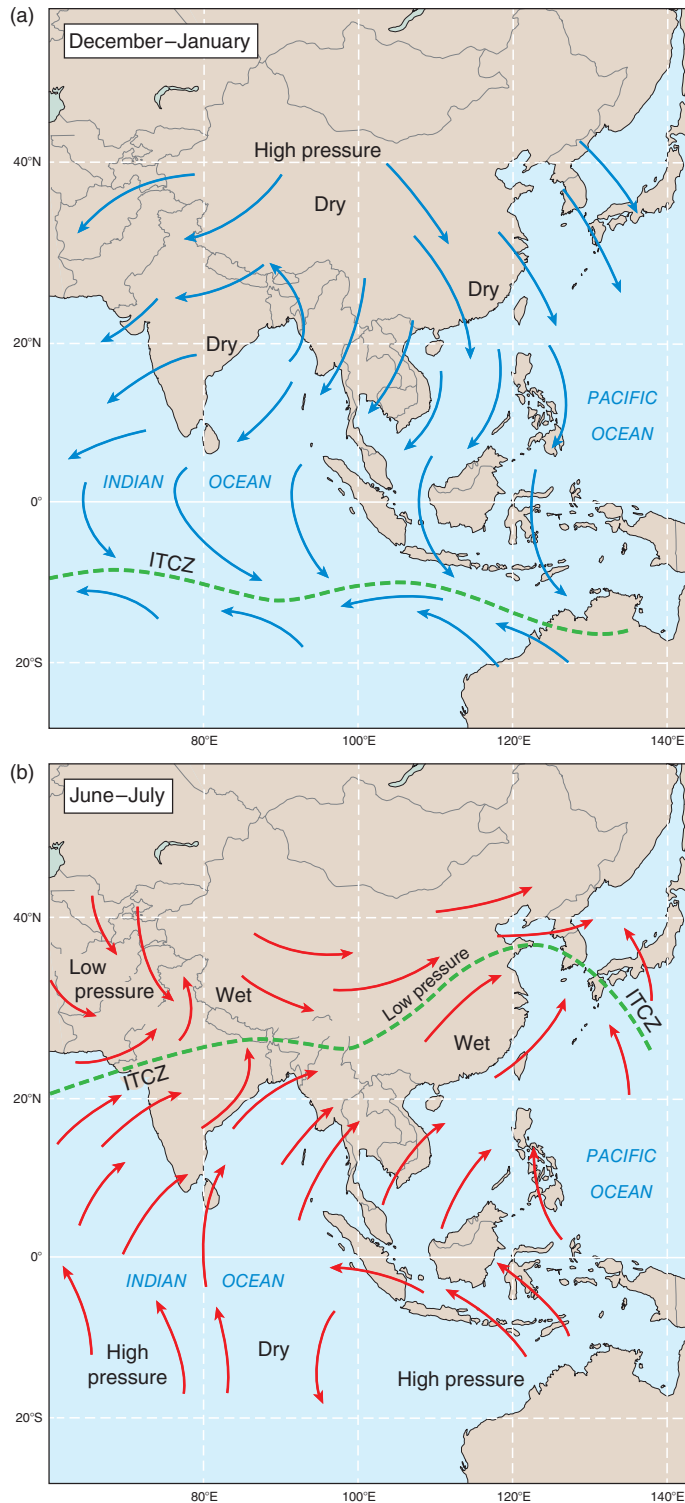


FIGURE 3.11 South Asian circulation patterns during the boreal winter and summer

The arrows indicate the dominant direction of the winds in the (a) winter and (b) summer. The monsoon precipitation develops in the summer, when the winds blow off the ocean and onto the continent and bring warm, moist air that feeds the monsoon convection. In the winter, high pressures develop over the continent and the winds flow off the land and over the ocean. Note the changing position of the Intertropical Convergence Zone with the season. (Redrawn from British Geographer, http://thebritishgeographer.weebly.com/uploads/1/1/8/1/11812015/3503731_orig.png)

north along the Horn of Africa and east around the tip of India. In the boreal winter, the contrast between land and ocean reverses because the ocean is warmer than the land. The air over the Himalayas is cool and dry and flows off the mountains, bringing with it dry conditions to India and adjacent regions. These winds also blow out across the ocean, reversing the ocean currents.

The basis of the overall seasonal process of the monsoon is familiar to anyone who has sat on a beach and experienced the daily progression of the sea breeze. The land heats up faster than the ocean during the day, creating a low-pressure region over the land and a high-pressure region over the ocean, thus causing the winds to blow from the ocean onto the land. In the evening, however, the land cools off more quickly than the ocean and the reversal of the pressure difference causes the winds to change direction, blowing from the land out to sea.

While these basic descriptions of the ocean–atmosphere (and land) interactions are sufficient for understanding the mean conditions of the South Asian monsoon, they are insufficient for understanding its spatial extent on land, the year-to-year variability in its strength, or the reason it collapses entirely. In particular, the year-to-year differences in the land temperature over India seem to have no relation to these factors. Nonetheless, several important features do stand out. First, variations in SST are in some ways diagnostic. As discussed earlier, El Niño tends to weaken the monsoon, and La Niña tends to strengthen it. The east–west SST contrast across the Indian Ocean also has been found to influence the strength of the monsoon. A second important influence is the Tibetan Plateau. Because of its high elevation, the Tibetan Plateau effectively heats air high into the troposphere. The extent of that heating determines the height of the north–south temperature difference in the atmosphere, which, in turn, influences the seasonal onset and intensity of the monsoon. The Tibetan Plateau also directs westerly flows of air to either its north or its south. The westerly air flow across Eurasia is cooler and drier than the monsoonal flow from the south. Therefore, when Eurasian air flows to the south of the Tibetan Plateau and mixes with monsoonal air, it reduces the temperature, moisture content, and convective height of the latter, acting to diminish the strength of the monsoon and limit its northward extent. While these features are part of a modern theory of how the monsoon changes over time, our understanding of the Asian monsoon remains incomplete.

Monsoons are, of course, not unique to South Asia. Other monsoon systems bring seasonal rains to sub-Saharan Africa, northern Australia, the southwestern United States and northern Mexico, and southern Brazil. These monsoons are driven by fundamental processes that are similar to those that underlie the South Asian monsoon, but each is subject to regional particulars associated with land topography, adjacent ocean conditions, and the geography of the continents on which they occur. Among these systems, the West African monsoon is particularly important because it profoundly influences the ecosystem and the livelihoods of peoples of the Sahel.

Given the importance of monsoon systems to agriculture and the general livelihoods of the people who are affected by them, the influence of global warming on their behavior is of significant concern. In general, enhanced warming over land areas, which is a major feature of global warming, might be expected to intensify the monsoons or lengthen their seasons. Similarly, rising SSTs in adjacent ocean regions might increase the amount of moisture that

feeds monsoonal systems. Despite these expectations, however, the warming that characterized the latter half of the twentieth century was accompanied by a decrease in the monsoonal rainfall in South Asia. At present, climate models provide little insight into how continued warming may influence the monsoons, with some showing a strengthening and others a weakening of the South Asian monsoon. Paleoclimate records nevertheless link the strength of the South Asian monsoon to large-scale changes in climate; a similar linkage has been established for the African monsoon. It therefore is unclear how the monsoon systems will respond to climate change, despite widespread evidence that suggests the systems are indeed sensitive to changing climate conditions.

THE ARCTIC OSCILLATION AND NORTH ATLANTIC OSCILLATION

As we discussed in chapter 1, a high-pressure zone develops at the surface in the high latitudes of the Northern Hemisphere due to the limited solar radiation received over the Arctic and the consistently cold conditions that exist there as a consequence. The strength and location of the high pressure fluctuate, however, and impose variability on mid-latitude weather. The phenomenon associated with these oscillations in pressure is known as the *Arctic Oscillation* (AO) or the *Northern Annular Mode*.

What are the consequences of these oscillations? When the pressure over the Arctic is lower than normal, it reflects a stronger gradient in temperature between the polar and equatorial regions. This enhanced gradient strengthens the polar jet stream, which, in turn, creates a stronger barrier between the cold Arctic air to the north and the warmer air to the south. This helps to confine the cold-air masses to the Arctic and limits cold-air outbreaks in the mid-latitudes that can cause strong winter storms. In contrast, if the pressure over the Arctic is higher than normal, the jet stream weakens and tends to bend and wiggle more. The bends and wiggles allow more cold-air outbreaks that propagate southward and establish strong mid-latitude storms. For instance, the AO was responsible for multiple extreme weather outbreaks in the United States during the boreal winters of 2009–2010 and 2010–2011. These were colloquially attributed by the popular press to the “Polar Vortex,” a phenomenon that occurs in both hemispheres and is associated with the same large-scale circulation of cold air related to the AO. The Polar Vortex is nevertheless indicative of those circulations higher up in the atmosphere, typically the stratosphere, while the AO describes fluctuations in the surface expression of those circulations. We shall revisit the AO in the context of how it affects Arctic sea ice (chapter 9).

A phenomenon tied to the same pressure fluctuations associated with the AO is the *North Atlantic Oscillation* (NAO), which is defined by an oscillation of atmospheric pressure between the polar and subtropical North Atlantic. The AO and NAO are specific descriptions of the same general atmospheric phenomenon: the variations of north–south pressure gradients in the high northern latitudes. The AO describes the variations on a larger hemispheric scale, while the NAO describes them over the North Atlantic sector. Specifically, a permanent low-pressure system is located over Iceland and the surrounding region of the North Atlantic, while a predominantly high-pressure system is located over the Azores and the surrounding region of the subtropical Atlantic. The strength and locations of these

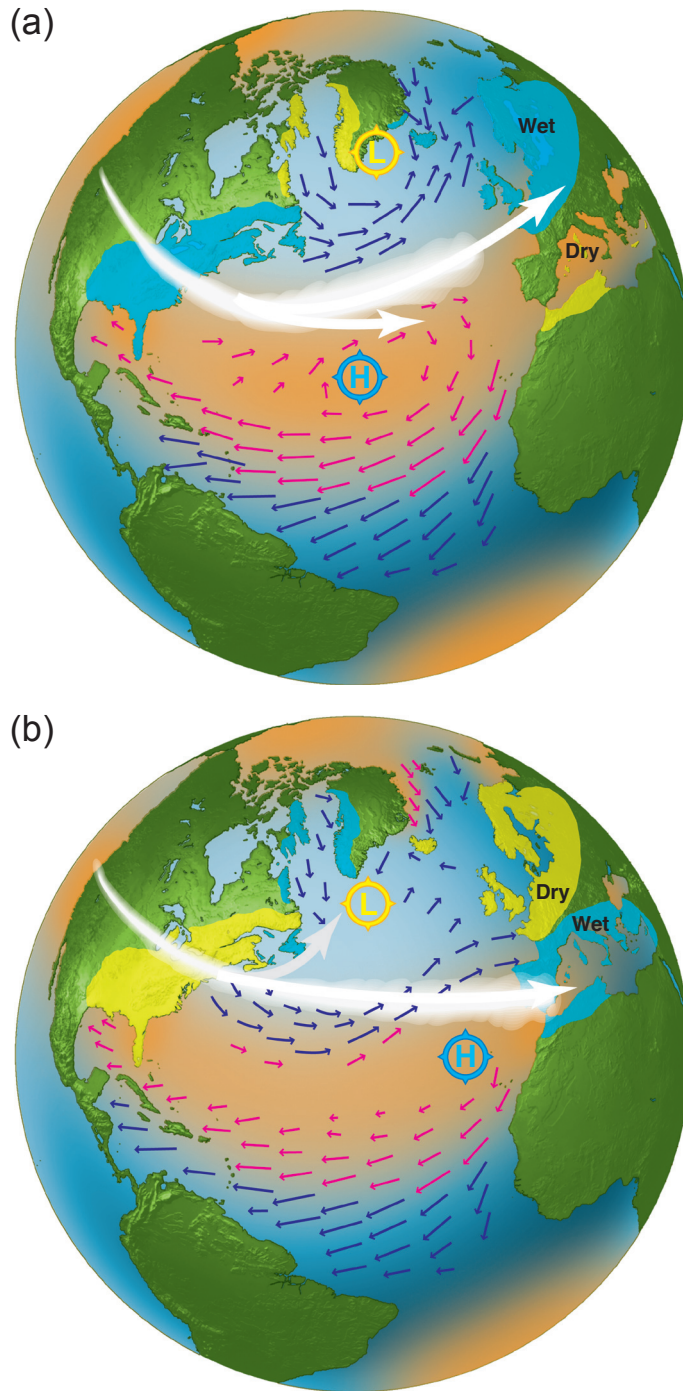


FIGURE 3.12 The North Atlantic Oscillation

The North Atlantic Oscillation is defined by an index that is typically expressed as the difference in atmospheric pressure between the Azores in the south and Iceland in the north. During the NAO's positive phase (a), the pressure is low over Iceland, enhancing the strength of the jet stream. This creates a stronger barrier between the cold Arctic air to the north and the warmer air to the south. Winter storms cross the Atlantic Ocean on a relatively northerly track, bringing relatively warm and wet winters to northern Europe and eastern North America. During NAO's negative phase (b), the storm tracks are farther south, bringing cooler and drier winters to northern Europe and eastern North America and wetter winters to the Mediterranean region. (Redrawn from M. Visbeck, "North Atlantic Oscillation," Lamont-Doherty Earth Observatory, Columbia University, <http://www.ldeo.columbia.edu/NAO/>)

pressure regions vary, however. They are measured using an NAO index that is typically expressed as the difference in atmospheric pressure between the Azores to the south and Iceland in the north.¹³ During the positive phase of the NAO, the mean state is characterized by relatively high atmospheric pressure centered over the Azores and relatively low pressure near Iceland, making the difference between the two pressure centers relatively large (figure 3.12). This state, which is most pronounced in the winter, results in strong winds transporting relatively warm, moist air from the Atlantic Ocean to northern Europe, Scandinavia, and Asia, bringing warm and wet winters to these regions. Unaffected by these winds, southern Europe and the Mediterranean basin remain relatively dry. The strong winds also create circulation patterns that draw cold, dry polar air into northeastern Canada and western Greenland and warm, moist air from the subtropical western Atlantic to the east coast of North America, causing the winters there to be unusually warm and wet. During the negative phase of the NAO, both the Azores high and the Iceland low are less intense and are displaced farther south. The lower pressure gradient between these two centers results in fewer and weaker storms that tend to follow a more east–west track, bringing moisture to southern Europe and the Mediterranean region. Meanwhile, northern Europe and the east coast of North America are colder than normal.

The AO and NAO fluctuate over the course of a few weeks, but they also display decadal oscillations. Both measures therefore relate to important short-term influences on the weather and longer-term influences on marine and terrestrial ecosystems, water supply, energy consumption, agriculture, and other systems. For example, by bringing warmth to regions in the North Atlantic, the positive phase of the NAO that persisted for the last several decades of the twentieth century had a negative effect on locations already suffering from diminished cod fisheries (figure 3.13).¹⁴ It also brought drought conditions to the Mediterranean, reducing the supply of drinking water and negatively affecting the harvests of olives, grapes, and other commodities.

Understanding why the AO and NAO oscillate on longer timescales remains elusive, and multiple factors have been suggested as contributing to their decadal variability. Specifically regarding the NAO, these factors include ocean influences as far afield as the tropics and more local impacts originating in the North Atlantic. The extent of sea ice in the Arctic Ocean, solar variability, and the amount of water vapor in the stratosphere have also been suggested as having an impact on the longer-term state of the AO and NAO. Further understanding the role of these factors will be important for possibly predicting the decadal fluctuations in the AO and NAO, as well as understanding how global warming will affect them.

THE PACIFIC DECADEAL OSCILLATION AND ATLANTIC MULTIDECADEAL OSCILLATION

While the AO and NAO describe atmospheric-pressure variations that oscillate over both weeks and decades, other oscillations associated with ocean conditions can act over decades or longer. Two important examples are the *Pacific Decadal Oscillation* (PDO) and the *Atlantic Multidecadal Oscillation* (AMO). Both the PDO and the AMO are associated with warming and cooling phases of SSTs in the respective Pacific and Atlantic Ocean basins. The

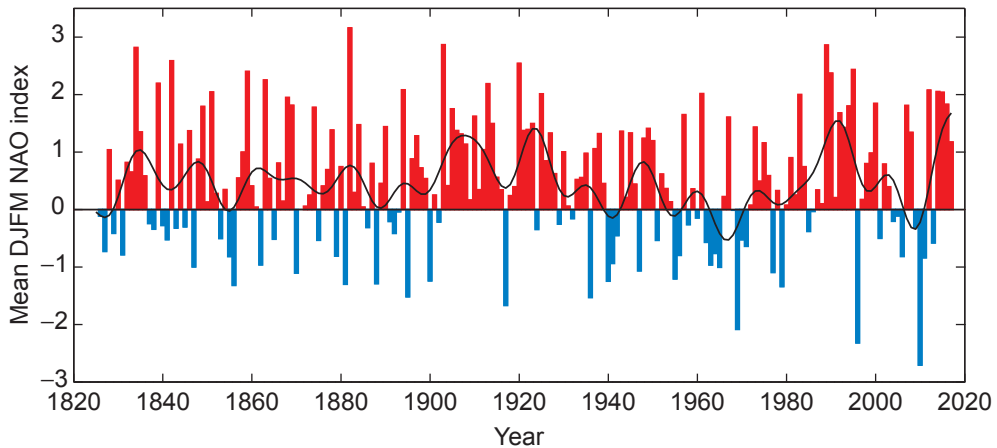


FIGURE 3.13 The average December–March value of the NAO index, 1825–2017

The phase of the NAO varies annually, while the black line represents the decadal smoothed average and shows the variability in the NAO over longer timescales. (Data from KMNI Climate Explorer, “North Atlantic Oscillation Index,” <https://climexp.knmi.nl/data/inao.dat>)

PDO is centered over a region that is north of 20°N , while the AMO is a pattern that occurs over the whole of the Atlantic Ocean in the Northern Hemisphere. During the positive phase of the PDO, the region over the western Pacific is cold, while a section of the eastern Pacific is warm; this pattern reverses itself during the negative phase of the PDO (figure 3.14). For the AMO, the positive phase is associated with warm SSTs throughout the North Atlantic basin, while in the negative phase colder temperatures dominate (figure 3.15).¹⁵ Fluctuations between the positive and negative states of both the PDO and the AMO take decades, although the AMO oscillations occur over longer timescales than do the PDO variations. Indeed, it can be seen that a single oscillation in the AMO took most of the twentieth century to complete.

As we have discussed throughout this chapter, the ocean and the atmosphere interact to cause changes in each other, and both the PDO and the AMO are additional examples of how this happens. Both oscillations are ocean phenomena, but they influence the atmosphere and weather. Through teleconnections, the positive phase of the PDO brings above-normal winter temperatures to the west coast of North America from Alaska down to Washington and Oregon, while bringing below-normal winter temperatures to the American Southwest and Mexico. The positive phase of the PDO is also associated with drought over the northern United States, while its negative phase is tied to drought in the Southwest. Over the Asian monsoon region, the negative phase of the PDO causes increased summer precipitation and lower temperatures, while the positive phase triggers the opposite. The PDO also affects salmon production in the Northeast Pacific: populations tend to grow during the negative phase of the PDO, while they decline during the positive phase. It was indeed this effect that first led to the identification of the PDO as an important oscillation in the northern Pacific.¹⁶ For the AMO, its negative phase brings drought to the African Sahel, as well as lower temperatures and reduced precipitation in Europe. During its positive phase, droughts tend to occur in North America, and the incidence of tropical hurricanes in the

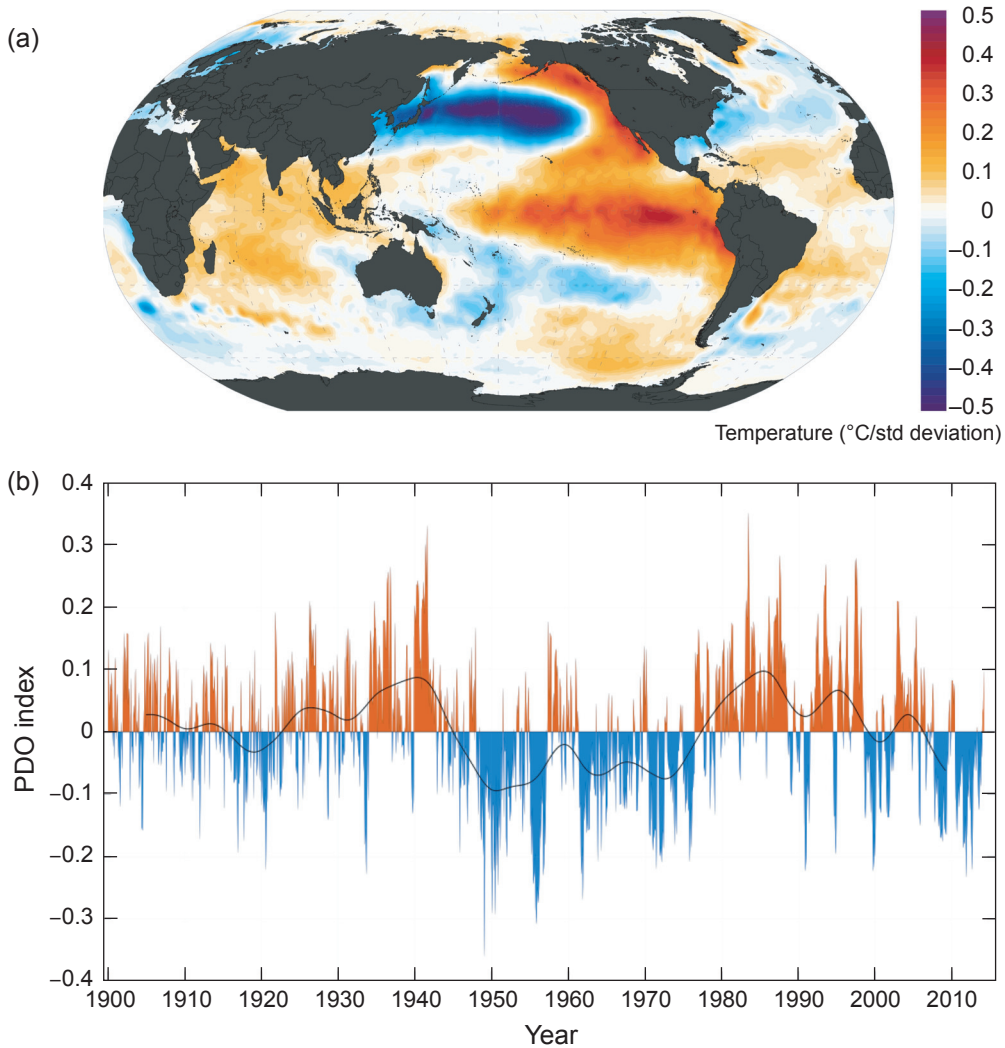


FIGURE 3.14 The Pacific Decadal Oscillation

(a) The geographic distribution of the positive phase of the Pacific Decadal Oscillation. Temperature is recorded in terms of °C per standard deviation of the PDO index. (b) The distinct oscillations over several decades between its positive and its negative phases. The black line is the 121-month smoothed trend. ([a] Giorgiopp2, “PDO Global Pattern,” March 19, 2012, CC BY-SA 3.0, <https://commons.wikimedia.org/w/index.php?curid=18754081>; [b] Giorgiopp2, “Monthly Values for the Pacific Decadal Oscillation Index,” CC BY-SA 3.0, <https://commons.wikimedia.org/w/index.php?curid=13297650>)

Atlantic Ocean increases, while they decrease during the negative phase (figure 3.16). The AMO is also connected to sea-level fluctuations in the Atlantic basin.¹⁷

Because of the length of the oscillatory cycles of the PDO and AMO, we have few cycles in the 100 to 150 years of the observational record with which to characterize their variability. As a result, the mechanisms that cause the two oscillations are not well understood. Thus the PDO and AMO are good examples of how other scientific approaches—including paleoclimate research and climate-modeling studies—are necessary to understand phenomena that are poorly observed over the instrumental record.

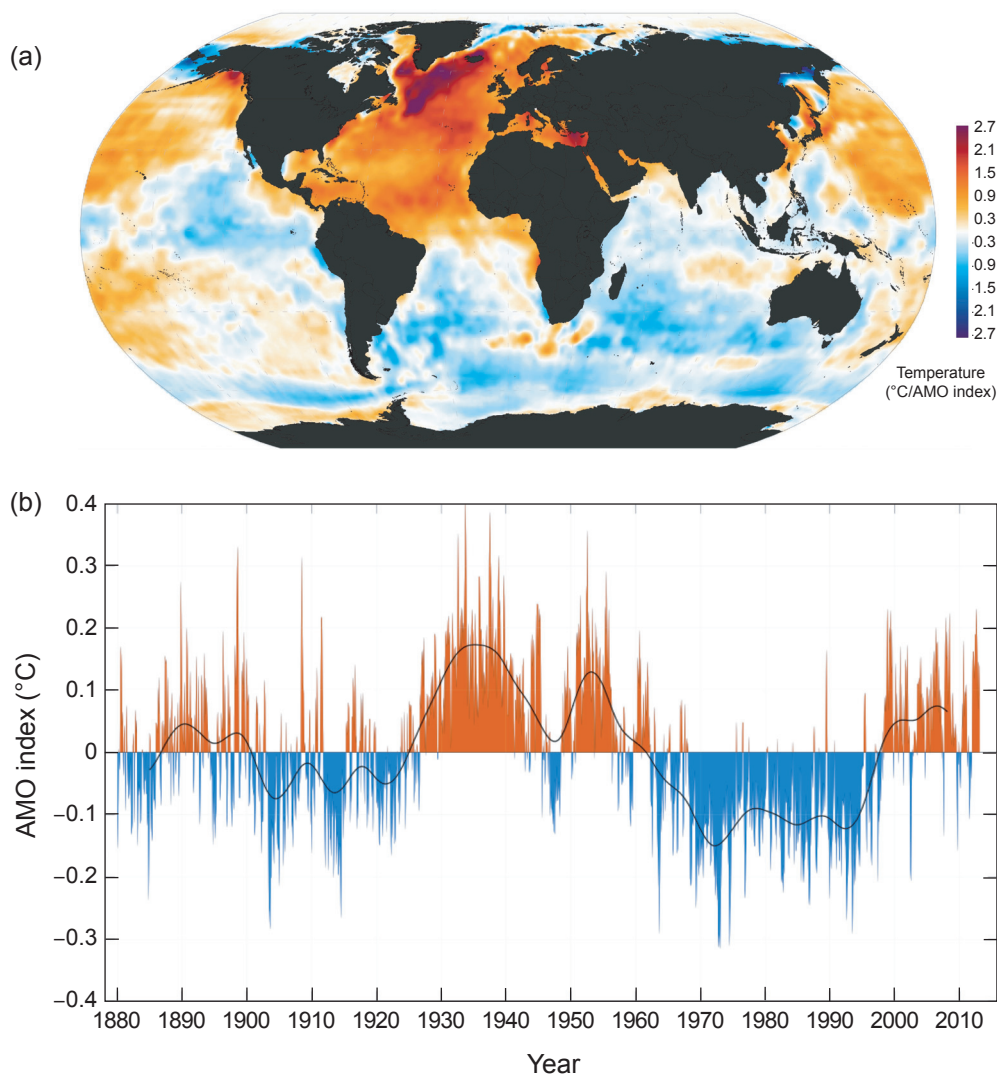


FIGURE 3.15 *The Atlantic Multidecadal Oscillation*

(a) The geographic distribution of the positive phase of the Atlantic Multidecadal Oscillation. Temperature is expressed as the monthly temperature anomaly divided by the AMO index temperature. (b) The distinct oscillations over several decades between its positive and its negative phases. The black line is the 121-month smoothed trend. ([a] Giorgiopp2, “Atlantic Multidecadal Oscillation,” April 19, 2014, CC BY-SA 3.0, <https://commons.wikimedia.org/w/index.php?curid=32266213>; [b] Giorgiopp2, “Atlantic Multidecadal Oscillation,” March 17, 2012, CC BY-SA 3.0, <https://commons.wikimedia.org/w/index.php?curid=18733673>)

THE IMPACT OF ONE OSCILLATORY MODE ON ANOTHER

A final word is necessary about how oscillations may affect each other. While ocean–atmosphere interactions give rise to individual oscillations, this coupling also allows the state of one oscillatory system to influence another. These impacts are often complicated and are not fully understood, but several examples illustrate how different oscillatory modes are tied together. ENSO, for instance, can affect the state of the NAO through teleconnections

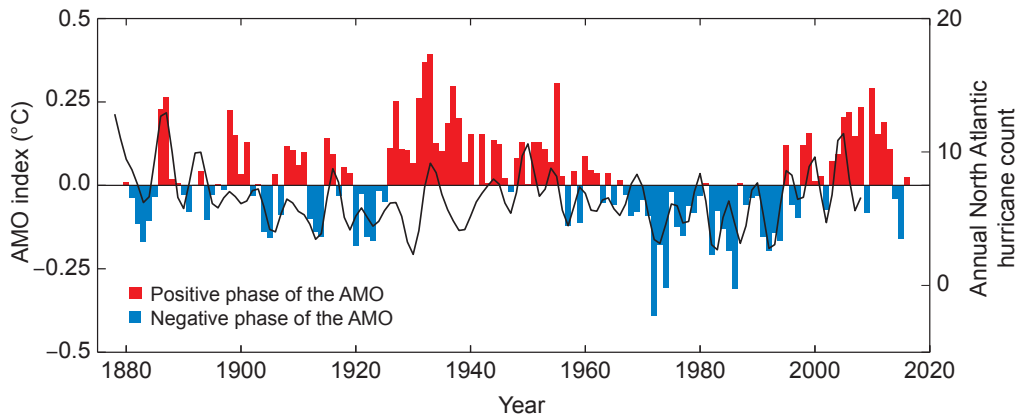


FIGURE 3.16 *The relationship between the AMO and the hurricane count in the North Atlantic Ocean*

During the positive phase of the AMO, the number of hurricanes increases, while the count goes down during the negative AMO phase. The black line is the annual number of hurricanes in the North Atlantic. (Data from KMNI Climate Explorer, “AMO Index,” https://climexp.knmi.nl/data/iamo_hadsst_ts.dat, and “Adjusted Number of Hurricane Days over the North Atlantic,” the latter based on Vecchi and Knutson, “Estimating Annual Numbers of Atlantic Hurricanes Missing from the HURDAT Database [1878–1965] Using Ship Track Density”)

that travel through the stratosphere.¹⁸ The NAO can have an impact on the South Asian monsoon by altering the amount of snow that falls over the Eurasian continent and, consequently, the amount of heating in the troposphere over that region.¹⁹ As we have noted, ENSO also affects the South Asian monsoon by weakening the Walker circulation during El Niños and, consequently, reducing the strength of the monsoon (the opposite is true during La Niñas). While the NAO can change the state of the Atlantic on subannual and interannual timescales, the AMO can influence the state of the NAO over decades. In particular, a positive AMO tends to be associated with more frequent negative winter NAO events, and thus colder wintertime temperatures over Europe and the eastern United States.²⁰ The AMO, perhaps through its impacts on the NAO, can also influence the variability of the South Asian monsoon.²¹ Each of these interactions illustrates how the ocean and the atmosphere interact beyond single oscillatory modes and can therefore create a rich range of variability as multiple oscillations interact in complex ways.

KEY POINTS IN THIS CHAPTER

The important connections between the ocean and the atmosphere should now be evident. Although we only touched on the role of the land surface, it too couples with the atmosphere and the ocean and plays an important role in climate variability and change. These collective interactions are manifest in a variety of ways. As examples, we encountered ENSO, the monsoons, the AO and NAO, and the PDO and AMO—all phenomena that oscillate on increasingly longer timescales and contribute to hemispheric and global weather and climate fluctuations through teleconnections. Among these, ENSO stands out for its global reach and, next to the seasons, represents the most important recurrent change in climate. There are additional oscillations that we have not considered, including the Indian Ocean Dipole, the

Southern Annular Mode, and the Madden-Julian Oscillation. Each contributes to the natural variability of the climate system and has impacts on regional human societies and ecosystems. Some can be modeled and partially predicted on different timescales, while others cannot. Several also interact with each other in complex ways. Global warming is likely affecting these natural climate rhythms, but separating the effects of rising atmospheric greenhouse gases from the processes that create the natural oscillations remains a challenge.

To summarize what we have learned:

1. Because of their density contrast, the ocean and the atmosphere do not mix in any significant way. Thus they interact through exchanges of energy, mass, and momentum at the ocean–atmosphere boundary.
2. The quintessential ocean–atmosphere phenomenon is the El Niño–Southern Oscillation (ENSO), the result of coupled ocean–atmosphere dynamics across the tropical Pacific Ocean.
3. During the neutral state of ENSO, the trade winds converge on the equator across the Pacific Ocean and cause warm water to collect in the Western Warm Pool (WWP), while upwelling brings colder water to the ocean surface in the east. This causes an east–west sea-surface temperature (SST) gradient that helps to establish low atmospheric pressure in the west and high pressure in the east, enhancing the easterly winds. Strong tropical convection over the WWP yields precipitation, while the upper-troposphere return flow descends in the east, adding to the high pressure there and completing the Walker circulation.
4. During El Niño, the trade winds diminish, and the water in the WWP flows back to the east. The center of atmospheric convection moves to the central and eastern Pacific Ocean and brings precipitation to those regions. Drought exists over the western Pacific, including parts of Australia and Indonesia.
5. During La Niña, the neutral conditions in the tropical Pacific Ocean strengthen and the east–west SST contrast in the tropical Pacific becomes larger. Enhanced precipitation occurs in the western Pacific, while drier and colder conditions exist over the eastern Pacific.
6. Both El Niño and La Niña bring dramatic changes in temperature and rainfall to a variety of regions around the globe through teleconnections. They also influence severe weather, such as the incidence of hurricanes in the Atlantic Ocean and their intensity in the Pacific.
7. The state of ENSO is characterized using the Southern Oscillation Index (SOI) and Niño indexes, but it is also measured in more detail using instruments on satellites, ocean buoys, ships, and radiosondes. These measurements are used in mathematical models that have proved to be reasonably good predictors of El Niño and La Niña six or so months in advance.
8. Monsoons affect multiple regions around the globe. They are fundamentally a consequence of the differential heating of land and ocean surfaces. The temperature contrast develops in the summer, when the land heats up more than the surrounding ocean. Monsoons bring rain to billions of people, who depend on their moisture delivery for agriculture and other water resources.

9. The Arctic Oscillation (AO) and North Atlantic Oscillation (NAO) are measures of the strength of the high-pressure region that occurs over the high northern latitudes in the winter. As this pressure varies, so does the strength and stability of the mid-latitude jet stream, which introduces important fluctuations in weather. The AO and NAO vary on daily timescales, but their annual average conditions display variations on decadal timescales as well.

10. The Pacific Decadal Oscillation (PDO) is a warming and cooling of SSTs in the Pacific Ocean north of 20°N. During the positive phase of the PDO, temperatures over the central and eastern part of the northern Pacific are cold, while temperatures along the west coast of North America are warm. This pattern reverses during the negative phase of the PDO.

11. The Atlantic Multidecadal Oscillation (AMO) is a warming and cooling of SSTs over the whole of the Atlantic Ocean basin in the Northern Hemisphere. The entire cycle appears to occur over four to eight decades, although the climate record of the AMO is too short to precisely characterize it.